Building quantum codes with transversal CCZ

from projective geometry and SAT solvers

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Joint work with Ken Brown

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Outline

1. **Quantum error correction and transversal gates**
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Pauli strings and stabilizer codes

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. For $a, b \in \mathbb{F}_2^n$, the *Pauli strings*

$$X(a) = X^{a_1} \otimes \cdots \otimes X^{a_n}, \qquad Z(b) = Z^{b_1} \otimes \cdots \otimes Z^{b_n}$$

act by $X(a)|c\rangle = |c \oplus a\rangle$ and $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$.

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- $[\![n, k, d]\!]$: the code has dimension 2^k, and every Pauli string acting on fewer than d qubits is detected or acts as a scalar on the code.

CSS codes

For binary linear codes $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] is the stabilizer code with stabilizers $X(a)$, $a \in \mathcal{C}_2$, and $Z(b)$, $b \in \mathcal{C}_1^\perp$. They commute because $a \cdot b = 0$. Write $\mathcal{C}_2 = \operatorname{rowspan} S$ and $\mathcal{C}_1 = \operatorname{rowspan} \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$, with stabilizer rows S $(s \times n)$ and logical rows K $(k \times n)$. The logical basis is

$$|x\rangle_L = 2^{-s/2} \sum_{y \in \mathbb{F}_2^s} |xK \oplus yS\rangle, \qquad x \in \mathbb{F}_2^k.$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
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- $Z(e)$ has syndrome Se^T and is undetected if and only if $e \in \mathcal{C}_2^\perp$. $X(e)$ is undetected if and only if $e \in \mathcal{C}_1$. So decoding uses the binary codes $\mathcal{C}_2^\perp$ and $\mathcal{C}_1$.

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- Logical operators: $X(a)$, $a \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and $Z(b)$, $b \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$. With $d_X = \min \operatorname{wt}(\mathcal{C}_1 \setminus \mathcal{C}_2)$ and $d_Z = \min \operatorname{wt}(\mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp)$, the code is $[\![n, k, d]\!]$ with $k = \dim \mathcal{C}_1 - \dim \mathcal{C}_2$ and $d = \min(d_X, d_Z)$.

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Transversal gates and their limits

A logical gate $\bar{U}$ is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space. A fault in one U_j remains a single-qubit error.

$$H = \tfrac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}), \quad \mathrm{CCZ}|x_1 x_2 x_3\rangle = (-1)^{x_1 x_2 x_3}|x_1 x_2 x_3\rangle.$$

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- Magic-state distillation [BK05] turns many noisy copies of a non-Clifford state into fewer accurate copies, using Clifford gates. Each copy implements one non-Clifford gate.

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Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode $[15,4,8]$	$T^{\otimes 15}$ acts as logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	A layer of T and $T^\dagger$ acts as logical CCZ.

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CNOT between two blocks is transversal for every CSS code. Since Clifford $+T$ is universal, [EK09] rules out the missing gates.

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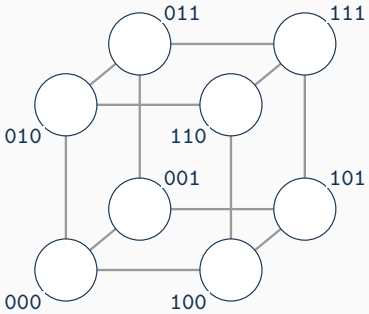
Question: which CSS codes with $k = 3$ have a choice $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ acting as logical CCZ?

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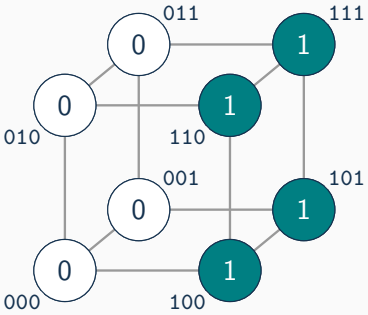
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$, given by its values at the points. Its weight is $\mathrm{wt}(c) = |c^{-1}(1)|$. The overlap $c \wedge c'$ is 1 where both vectors are 1, so it is the vector of the product function cc'.

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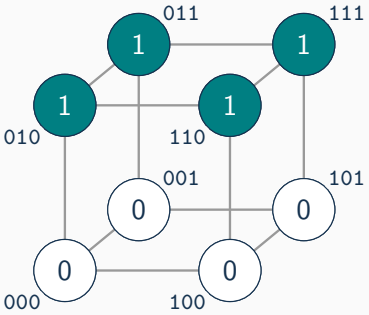
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Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

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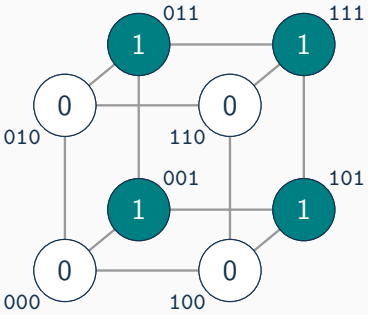
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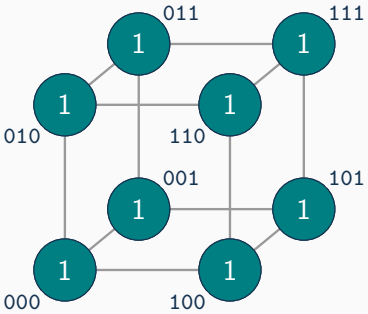
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The constant vector $\mathbf{1}$.

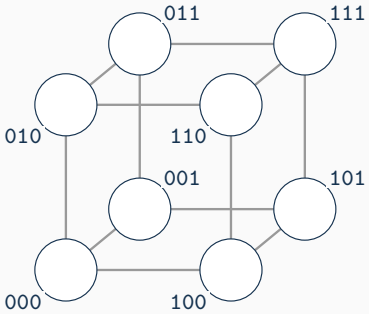
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$\mathrm{RM}(r, m)$ is the code of the vectors given by polynomials of degree at most r. For example, $\mathrm{RM}(1, 3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$ is an $[8, 4, 4]$ code.

The cube code $[\![8, 3, 2]\!]$

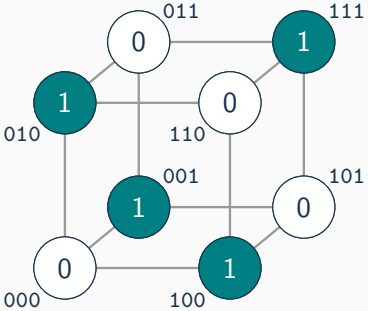


Take $S = \mathbf{1}$, so the only stabilizer generator is $X(\mathbf{1}) = X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$xK = x \cdot u = x_1u_1 \oplus x_2u_2 \oplus x_3u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

The cube code $[\![8, 3, 2]\!]$



The codeword $x \cdot u$ for $x = 111$.

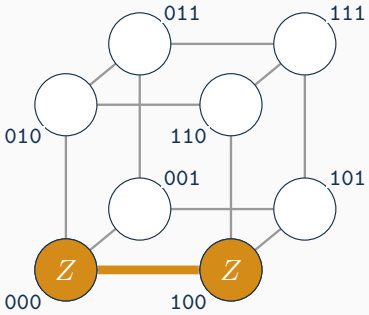
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For $x = 111$ the codeword $x \cdot u$ is $u_1 \oplus u_2 \oplus u_3$.

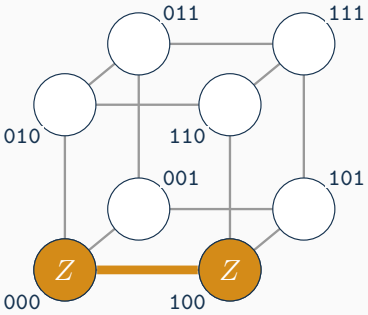
Distances of the cube code



Z on the qubits 000 and 100.

Let $\mathbf{1}_u$ be the vector with a single 1 at u. Z on the qubits 000 and 100 is $Z(e)$ with $e = \mathbf{1}_{000} \oplus \mathbf{1}_{100}$.

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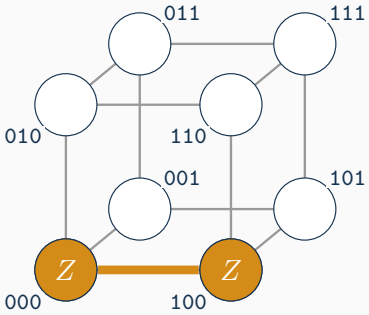


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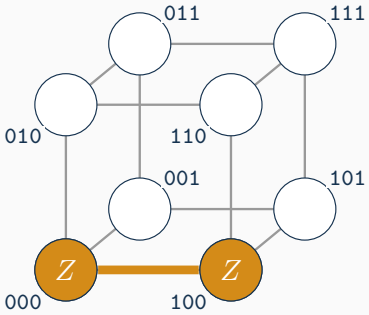


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- Every codeword is $c = b \oplus x \cdot u$, with $c(000) = b$ and $c(100) = b \oplus x_1$. So $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.

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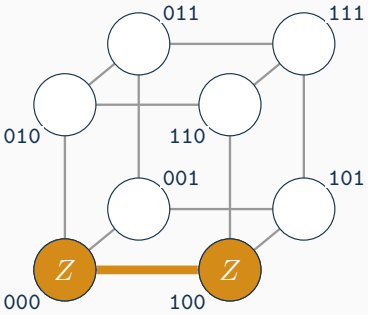


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- $e \cdot \mathbf{1} = 0$, so $Z(e)$ commutes with the stabilizer $X(\mathbf{1})$. Each $Z(\mathbf{1}_u)$ anticommutes with it, so $d_Z \geq 2$.
- Every codeword is $c = b \oplus x \cdot u$, with $c(000) = b$ and $c(100) = b \oplus x_1$. So $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.
- $\mathrm{RM}(1,3)$ has minimum weight 4, so $d_X = 4$.

Distances of the cube code



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This is the smallest interesting colour code: a transversal $T/T^\dagger$ layer gives logical CCZ [CH17].

Codeword weight, physical phase, and logical phase

For $b \in \{0,1\}$, $T^{\pm 1}|b\rangle = e^{\pm i\pi b/4}|b\rangle$. Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$. Then

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi W_\Gamma(c)/4}|c\rangle, \qquad W_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $W_\Gamma(c)$ counts the ones of c at T sites minus the ones at $T^\dagger$ sites.

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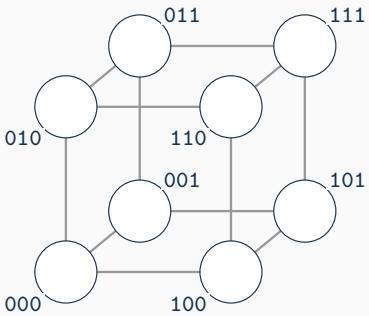
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The product acts as logical CCZ if and only if every codeword of the coset $xK \oplus \mathcal{C}_2$ receives the phase $(-1)^{x_1x_2x_3}$:

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

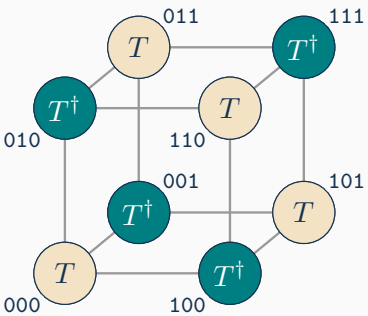
Choosing the signs on the cube



Every codeword of the cube code is an *affine function* $f_{x,b} = b \oplus x \cdot u$. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so $f_{x,b}$ has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$. It is the codeword $x \cdot u$ for $x = 111$.

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

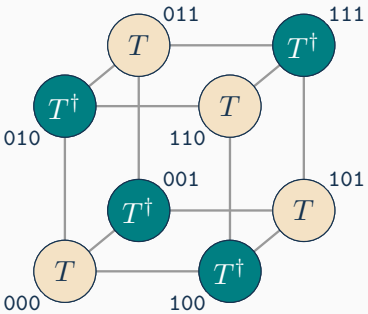
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Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. With the Hamming distance $d_H(p, c) = \mathrm{wt}(p \oplus c)$,

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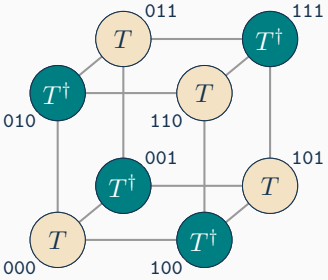
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$x \neq 111$: $p \oplus f_{x,b} = b \oplus (x \oplus 111) \cdot u$ is nonconstant, so $d_H = 4$ and $W_\Gamma = 0$.

$x = 111$: $f_{x,b} \in \{p, \mathbf{1} \oplus p\}$, so $d_H \in \{0, 8\}$ and $W_\Gamma = \pm 4$, phase -1.

Reading the cube phase from overlaps



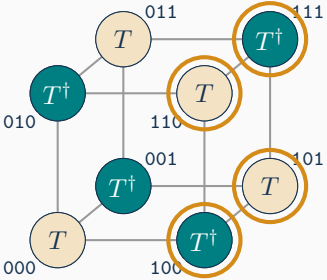
For bits a_1, a_2, a_3, XOR has the integer expansion

$$a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

For $c = x_1K_1 \oplus x_2K_2 \oplus x_3K_3$, apply it at each vertex u with $a_i = x_iK_i(u)$, weight by Γ_u, and sum:

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Reading the cube phase from overlaps



$K_1 = u_1$: the face $u_1 = 1$.

For bits a_1, a_2, a_3, XOR has the integer expansion

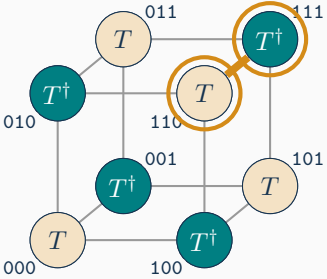
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The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $W_\Gamma(K_1) = 0$. Likewise for K_2, K_3.

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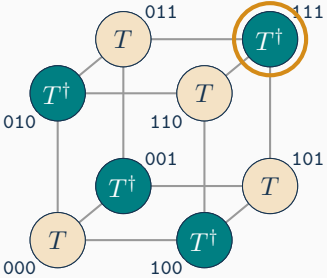
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The vertex 111 is a $T^\dagger$, so $W_\Gamma(K_1 \wedge K_2 \wedge K_3) = -1$ and $W_\Gamma(c) = -4x_1 x_2 x_3 \equiv 4x_1 x_2 x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

The same expansion for any number of binary vectors is the *Bravyi–Haah parity identity* [BH12]

$$W_\Gamma(a_1 \oplus \cdots \oplus a_m) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1} W_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Expand $W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$ with it and compare the coefficients of the monomials in (x, y). For distinct rows g_a, g_b, g_c of $G = [K; S]$:

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The $[\![8, 3, 2]\!]$ code satisfies all three: rows have $W_\Gamma = 0$, pairs of rows overlap in faces or edges with $W_\Gamma = 0$, and triples overlap in an edge ($W_\Gamma = 0$) or in the vertex 111 ($W_\Gamma = -1$).

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Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. **Triorthogonality and the CCZ conditions**
4. Projective geometry and the 48-qubit code
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6. Lower bounds and open questions

Historical context

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices, transversal T up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via RM indicators.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows label logical T outputs. Even-weight rows give X-type stabilizers.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j \equiv 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every vector in $\mathcal{H}$ has even weight.

Nezami–Haah classify the parent spaces $\mathcal{H}$, then obtain codes by puncturing coordinates. This work fixes S from geometry first, then searches for a compatible K.

Conditions for quasi-transversal CCZ

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) \equiv 0$
S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17]. They give CCZ up to a diagonal Clifford correction (*quasi-transversal*).

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A physical $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

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Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$Q_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- The stabilizer code is fixed by projective geometry: $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$.
- The logical rows come from a SAT search.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- We found codes at block lengths 48 to 496 using S constructed from $\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2)$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

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4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \implies \text{weight-one errors are detected}, \qquad \beta_i \neq \beta_j \implies \beta_i \oplus \beta_j \neq 0.$$

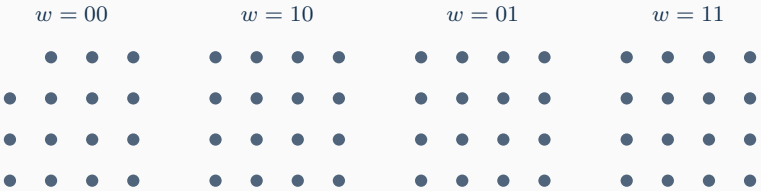
Distinct nonzero columns are points of $\mathrm{PG}(s-1,2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the stabilizer columns, independent of the choice of logical rows.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

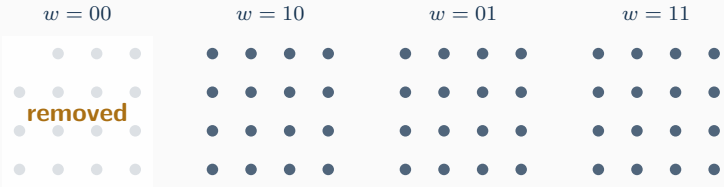
Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.



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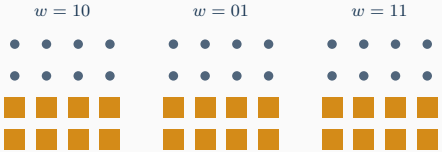


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ stabilizer code.

Hyperplane sections and codeword weights

A codeword of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



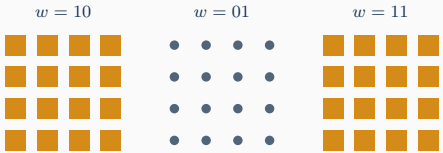
$\lambda \neq 0$, for example $\ell = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

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$\lambda = 0$, for example $\ell = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

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Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p \backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no valid code.

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3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. **Column indicators, logical rows, and phase signs**
6. Lower bounds and open questions

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

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For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, every x occurs with $y = 1$ and no column has $y = 0$. So $\chi_G(x, y) = 1$ exactly when $y = 1$, that is $\chi_G(x, y) = y$.

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Q_{48}: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

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Since $\mathrm{RM}(3, r)^\perp = \mathrm{RM}(r-4, r)$, and $y_1 \cdots y_s$ (one at the 8 points $(x, 1, \ldots, 1)$) is one solution [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

Logical rows in the column picture

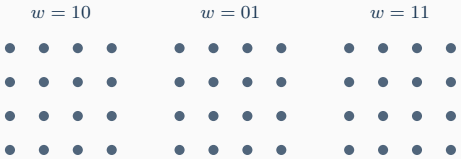
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. With $[P] = 1$ if P holds and 0 otherwise, the CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



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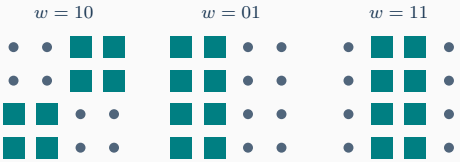
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S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 \oplus u_3$, $1 \oplus u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, \beta_l, \beta_l \beta_{l'}$ on $\mathcal{P}$. So $\kappa_i \in V_{\mathcal{P}} = \mathrm{ev}_{\mathcal{P}}(\mathrm{RM}(2, s))^{\perp}$, a linear condition.

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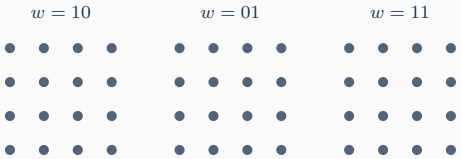
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If no pair extends, $\mathcal{P}$ is excluded. A SAT solver can also solve the full system at once. For Q_{48}, the columns at $u = 0000$ in the three blocks sum to 0 and their labels to $011 \oplus 110 \oplus 000 = 101$. These three Z errors form a logical operator, so $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$



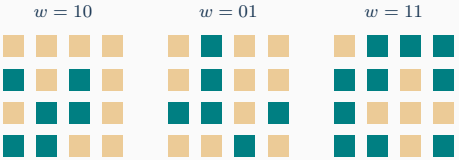
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For Q_{48} [LB26]:

$$\gamma(u, w) = u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u),$$
$$Q_1 = u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4,$$
$$Q_2 = u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.$$



Teal: $T^\dagger$ $(6, 6, 10)$. Gold: T. Total $26\,T + 22\,T^\dagger$.

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Stabilizer supports are Reed–Muller codewords

For projective S with support $\mathcal{P} \subset \mathbb{F}_2^s$, S1–S3 imply

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0 \pmod 2, \qquad 1 \le t \le 3.$$

Following [NH22], let $c = \chi_{\mathcal{P}} + \epsilon\delta_0$, where δ_0 is the indicator of the origin and ϵ makes $\mathrm{wt}(c)$ even. Every monomial of degree at least 1 vanishes at $\beta = 0$, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = \mathrm{wt}(c) \bmod 2$, which ϵ makes 0. Hence c is orthogonal to every monomial of degree at most 3, so $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

For Q_{48}: $c = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no physical $T/T^\dagger$ layer [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat, punctured at the origin. Q_{48} removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with circuit footprint 9.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $Q_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Our bound excludes $n \leq 38$, and [WJS26] excludes 39–46.

Open questions

- Which supports admit compatible labels and signs?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public